

Renato Paredes Venero

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PROFILE AND MAJOR ACHIEVEMENTS

I am a psychologist and computational cognitive neuroscientist specialising in neural network models of multisensory integration. My research combines experimental and computational methods to uncover the neural mechanisms driving alterations in Bodily Self-Consciousness across psychiatric conditions. I lead the development of the Scikit-NeuroMSI computational modelling framework and have published my theoretical and empirical findings in high-profile neuroscience and psychiatry journals. As an Assistant Professor at the Pontifical Catholic University of Peru, I founded and led the Computational Cognitive Neuroscience Lab, securing three internal grants and fostering international collaborations. Currently, as a Postdoctoral Researcher at the Max Planck Research Unit for Neurogenetics, I am expanding my methodological abilities to include computational neuroanatomy and 3D spatial analysis. My combined expertise in open-source software development, interdisciplinary collaboration, and computational modelling positions me to advance a mechanistic theory of the bodily self and drive translational impact in clinical research, robotics, and VR technologies.

RESEARCH EXPERIENCE

Max Planck Research Unit for Neurogenetics

Postdoctoral Researcher

Dec 2025 – Present

- I conduct computational neuroanatomy research of the mouse olfactory system. Focus on segmentation, registration and spatial analysis of 3D STPT images.

Research Internship

Dec 2024 – Apr 2025

- I conducted statistical analysis of STPT images of the mouse olfactory bulb, resulting in 1 peer-reviewed article.

Pontifical Catholic University of Peru

Head of Computational Cognitive Neuroscience Lab

Jan 2023 – Dec 2025

- I started the implementation of this new lab in the Department of Psychology, actively setting standards for workflow, equipment, human resources and software management.
- I led a team of 12 researchers, assistants, and students working on cognitive neuroscience projects using experimental and computational methods, resulting in 3 internal research grants, 1 results-based management fund and 3 peer-reviewed publications.

Co-researcher (Department of Engineering)

Apr 2021 – Apr 2023

- Project: "Teleoperated Mobile Robot for Mental Health Delivery in Patients with Infectious Diseases"

Research Assistant (Department of Psychology)

Mar 2016 – May 2018

- Project: "Neurocognitive study of perception of actions and emotions in the interaction between human beings and humanoid robots"

The University of Edinburgh

Remote Research Assistant (Computational Psychiatry Lab)

Sep 2019 – Aug 2021

- I started the collaboration with two laboratories of world-class universities (NYU & University of Chieti-Pescara).
- I led the process of writing and publishing two articles on neural networks of peri-personal space in high impact journals.

University of Waseda

Data Analysis Consultant (School of International Liberal Studies)

May – Sep 2018

- I conducted data analysis in Human-Robot Interaction (HRI) studies.
- I co-authored papers in relevant HRI journals and conferences.

TEACHING EXPERIENCE

Pontifical Catholic University of Peru

Assistant Professor (Department of Psychology)

Jan 2025 – Present

- I was in charge of developing a Cognitive Neuroscience postgraduate programme and establishing academic cooperation agreements.
- Currently on leave for postdoctoral research period.

Full-time Lecturer (Department of Psychology)

Aug 2022 – Dec 2024

- I coordinated the “Behavioural Neuroscience” and “Neuropsychology” courses at the School of Psychology.
- I gave the courses “Statistics and Mathematical Psychology” and “Research Methods for Psychology” at the Postgraduate School.
- I created the course “Introduction to Python for Psychological Research” offered at undergraduate, postgraduate and continuing education levels.

Adjunct Lecturer (Department of Psychology)

Jan 2021 – July 2022

Teaching Assistant (Department of Psychology)

Mar 2017 – Sep 2018

Neuromatch Academy Inc.

Teaching Assistant

Jun 2020 – Jul 2022

- I provided support to students learning computational neuroscience and mathematical modelling of brain activity.
- I reviewed didactic material for theoretical and practical learning sessions, which resulted in 1 peer-reviewed publication.

EDUCATION

National University of Cordoba

Doctor en Neurociencias (equivalent to Ph.D.)

Cordoba, Argentina

Jun 2021 - Apr 2025

Supervisors: Dr. Peggy Seriès and Dr. Pablo Barttfeld

Thesis: “Neural Network modelling of multisensory integration in patients with schizophrenia and autism spectrum disorder”

Defence date: 09/04/2025

Grade: Approved with Outstanding Mention - 10/10

The University of Edinburgh

MSc. in Cognitive Science

Edinburgh, UK

Sep 2018 - Aug 2019

Supervisor: Dr. Peggy Seriès

Dissertation: “Computational modelling of peripersonal space representation in schizophrenia spectrum disorders”

Grade: Approved with Distinction.

Cognitive Science Msc Dissertation Prize

Pontifical Catholic University of Peru

B.A. in Psychology, with Outstanding Mention

Lima, Peru

Mar 2010 - Jul 2016

Supervisor: Dr. Alex Davila

Thesis: “Activity of the mirror neuron system during perception and action of finger movements”

Grade: Approved with Outstanding Mention

CONFERENCES AND WORKSHOPS

- Organiser at Workshop on Experimental Psycholinguistic Methods for Underdescribed Languages (PUCP - UZH - NCCR Evolving Language) 2025.
- Oral presentation at XL Congreso Interamericano de Psicología (CIP) 2025.
- Scientific Committee at X Congreso Latinoamericano de Psicología de la Salud (ALAPSA) 2025.
- Oral presentation at Adriatica Summer School 2024.
- Poster presentation at Bernstein Conference 2021.
- Programme Committee and oral presentation at the International Workshop on Evaluation Methods Standardisation for Human-Robot Interaction (EMSHRI) 2017.

ACADEMIC SERVICE

- Ad hoc Reviewer, *Frontiers in Psychology* (2026).
- Ad hoc Reviewer, *IEEE Robotics and Automation Letters* (2025).
- Ad hoc Reviewer, *Scientific Reports* (2022).
- Ad hoc Reviewer, *Revista de Psicología (PUCP)* (2022-2025).
- Conference Reviewer, *ICSR* (2020).
- Conference Reviewer, *RO-MAN* (2017-2025).

INVITED SEMINARS AND TALKS

- “Understanding the Brain: Rodrigo Quián Quiroga in conversation with Joseph Crawford, Mari Fernández, and Renato Paredes”. Talk at Hay Festival (Lima, Peru). November 2025.
- “Neural Network modelling of multisensory integration differences in Schizophrenia and Autism Spectrum Disorder”. Workshop at Institute for Adaptive and Neural Computation, The University of Edinburgh (Edinburgh, UK). March 2025.
- “Neural Network modelling of multisensory integration differences in Schizophrenia and Autism Spectrum Disorder”. Lab Meeting at Laboratory of Cognitive Neuroscience, EPFL Campus Biotech (Geneva, Switzerland). February 2025.
- “Innovations from the Computational Cognitive Neuroscience Laboratory at PUCP”. Talk at 11th International Psychology Congress: Challenges for Psychology in a 4.0 Context, Interamerican Society of Psychology (SIP) (Lima, Peru). October 2024.

PRIZES AND AWARDS

- Seal of Excellence for the project proposal “Decoding Illusions of the Self: Neural Network Modelling of Bodily Self-Consciousness” submitted under the call MSCA Postdoctoral Fellowships 2025 (European Commission).
- Cognitive Science MSc Dissertation Prize 2019 (School of Informatics, The University of Edinburgh).
- Excellence Scholarship “President of the Republic” 2018 (PRONABEC, Peruvian Ministry of Education). Rank: 1st. Stipend: € 50,000.
- Best paper award in the category “Excellence in Engineering Education. IEEE Global Engineering Education Conference (EDUCON) 2017.
- Student Mobility Programme. Interuniversity Development Centre (CINDA). Stipend: € 4,000.

FUNDING RECEIVED

- Annual Research Project Grant (CAP) - Individual Category (November 2024 - November 2025).
Funder: Research Promotion Office, PUCP.
Award: € 12,500 (Grant N°: CAP-PI1220-2024).
Project: “Melodies of the mind: Deciphering the impact of music on the regulation of affective and cognitive states of university students using EEG and NLP”.
- Annual Research Project Grant (CAP) - Group Category (November 2023 - November 2025).
Funder: Research Promotion Office, PUCP.
Award: € 35,000 (Grant N°: CAP-PI1060-2023).
Project: “Towards the development of a neurocomputational profile of Peruvian women with symptoms of CPTSD”.
- Research Support Grant (FAI) (June - December 2023).
Funder: Research Promotion Office, PUCP.
Award: € 2,000 (Grant N°: FAI-085-2023).
Project: “Multisensory integration and auditory sensitivity in students with high abilities at a private university in Metropolitan Lima”.
- Results-Based Management Fund (August - December 2023).
Funder: Institutional Projects Office, PUCP.
Award: € 15,000.
Project: “Positioning of the Computational Cognitive Neuroscience Laboratory”.

SUPERVISING AND MENTORING

- School of Informatics, The University of Edinburgh: I co-supervised 3 BSc and 3 MSc Informatics students on computational neuroscience projects with Dr. Peggy Seriès. 2022-2026.
- School of Mathematics, Astronomy, Physics and Computing (FAMAF), National University of Cordoba: I co-supervised 1 BSc Computer Science student on a computational neuroscience project with Dr. Juan Cabral. 1 obtained the Outstanding Mention (10/10) in his defence. 2025-2026.
- School of Psychology, PUCP: I supervised 4 BA Psychology students in cognitive science projects. 1 obtained the Outstanding Mention in her defence. 2021-2025.

NOTE

Due to the scarcity of PhD holders in Peru, national regulations allow Master’s degree holders to assume teaching and research positions. This allowed me to progressively take on roles as Lecturer (2021-2024), Lab Coordinator (2023-2025) and Assistant Professor (2025–present) prior to PhD completion. The computational nature of my PhD project, conducted mainly remotely at the National University of Cordoba (Argentina), facilitated the parallel management of these roles. During this period, I used my expertise in computational neuroscience to lead significant academic and research initiatives in a context with limited neuroscience researchers.